2.20 Hyperbolic Functions: Integration

Worked Solutions

1 a)

$$\int \sinh x \, dx = \cosh x + c$$

1 b)

$$\int \cosh x \, dx = \sinh x + c$$

1 c)

$$\int \operatorname{sech}^2 x \, dx = \tanh x + c$$

1 d)

$$\int \sinh 3x \, dx = \frac{1}{3} \cosh 3x + c$$

1 e)

$$\int \cosh \frac{x}{2} \, dx = 2 \sinh \frac{x}{2} + c$$

1 f)

$$\int \operatorname{sech}^2 9x \, dx = \frac{1}{9} \tanh 9x + c$$

2)

$$\begin{aligned}
\int_{\ln 2}^{\ln 3} \sinh x \, dx &= [\cosh x]_{\ln 2}^{\ln 3} \\
&= \cosh(\ln 3) - \cosh(\ln 2) \\
&= \frac{e^{\ln 3} + e^{-\ln 3}}{2} - \frac{e^{\ln 2} + e^{-\ln 2}}{2} \\
&= \frac{3 + \frac{1}{3}}{2} - \frac{2 + \frac{1}{2}}{2} \\
&= \frac{5}{3} - \frac{5}{4} = \frac{5}{12}.
\end{aligned}$$

3)

$$\begin{aligned}\int_{\ln 3}^{\ln 4} \cosh 2x\, dx &= \left[\frac{1}{2}\sinh 2x\right]_{\ln 3}^{\ln 4} \\ &= \frac{1}{2}\sinh(2\ln 4) - \frac{1}{2}\sinh(2\ln 3) \\ &= \frac{1}{2}\sinh(\ln 16) - \frac{1}{2}\sinh(\ln 9) \\ &= \frac{16 - 1/16}{4} - \frac{9 - 1/9}{4} \\ &= \frac{255}{64} - \frac{20}{9} = \frac{1015}{576}.\end{aligned}$$

4)

$$\begin{aligned}\int_0^{\ln 2} (15\cosh 3x - 9\sinh 3x)\, dx &= [5\sinh 3x - 3\cosh 3x]_0^{\ln 2} \\ &= 5\sinh(3\ln 2) - 3\cosh(3\ln 2) - (5\sinh 0 - 3\cosh 0) \\ &= 5\sinh(\ln 8) - 3\cosh(\ln 8) + 3 \\ &= 5\frac{8 - 1/8}{2} - 3\frac{8 + 1/8}{2} + 3 \\ &= \frac{315}{16} - \frac{195}{16} + 3 = \frac{21}{2}.\end{aligned}$$

5)

$$\begin{aligned}\int_{-\ln 2}^{\ln 2} \operatorname{sech}^2 x\, dx &= [\tanh x]_{-\ln 2}^{\ln 2} \\ &= \tanh(\ln 2) - \tanh(-\ln 2) \\ &= \frac{2 - 1/2}{2 + 1/2} - \frac{1/2 - 2}{1/2 + 2} \\ &= \frac{3}{5} - \left(-\frac{3}{5}\right) = \frac{6}{5}.\end{aligned}$$

6)

$$\begin{aligned}\lim_{\delta x \to 0} \sum_{x=0}^{\ln 2} (8\cosh 4x - 5\sinh 3x)\, \delta x &= \int_0^{\ln 2} (8\cosh 4x - 5\sinh 3x)\, dx \\ &= \left[2\sinh 4x - \frac{5}{3}\cosh 3x\right]_0^{\ln 2} \\ &= 2\sinh(4\ln 2) - \frac{5}{3}\cosh(3\ln 2) + \frac{5}{3} \\ &= 2\sinh(\ln 16) - \frac{5}{3}\cosh(\ln 8) + \frac{5}{3} \\ &= 16 - \frac{1}{16} - \frac{5}{6}\left(8 + \frac{1}{8}\right) + \frac{5}{3} \\ &= \frac{65}{6}.\end{aligned}$$

7)

$$\begin{aligned}
\int_0^k (6\sinh 3x + 15\cosh 3x)\,dx &= 8 \\
[2\cosh 3x + 5\sinh 3x]_0^k &= 8 \\
2\cosh 3k + 5\sinh 3k - 2 &= 8 \\
2e^{3k} + 2e^{-3k} + 5e^{3k} - 5e^{-3k} - 4 &= 16 \\
7e^{3k} - 3e^{-3k} - 20 &= 0 \\
7e^{6k} - 20e^{3k} - 3 &= 0 \\
(7e^{3k} + 1)(e^{3k} - 3) &= 0.
\end{aligned}$$

Since $e^{3k} > 0$, $e^{3k} = -\frac{1}{7}$ is impossible. Thus $e^{3k} = 3$, so $3k = \ln 3$ and $k = \dfrac{1}{3}\ln 3$.

8)

$$\begin{aligned}
\int_{-\pi/4}^{\pi/4} (\sec^2 x - \operatorname{sech}^2 x)\,dx &= [\tan x - \tanh x]_{-\pi/4}^{\pi/4} \\
&= \left(1 - \frac{e^{\pi/4} - e^{-\pi/4}}{e^{\pi/4} + e^{-\pi/4}}\right) - \left(-1 - \frac{e^{-\pi/4} - e^{\pi/4}}{e^{-\pi/4} + e^{\pi/4}}\right) \\
&= 2 + \frac{2e^{-\pi/4} - 2e^{\pi/4}}{e^{\pi/4} + e^{-\pi/4}} \\
&= \frac{4e^{-\pi/4}}{e^{\pi/4} + e^{-\pi/4}} = \frac{4}{e^{\pi/2} + 1}.
\end{aligned}$$

As $e^{\pi} + 1 > e^{\pi/2} + 1$, we have $\dfrac{4}{e^{\pi/2} + 1} > \dfrac{4}{e^{\pi} + 1}$. Hence the required inequality follows.

9 a)

$$\int \sinh(2x + 5)\,dx = \frac{1}{2}\cosh(2x + 5) + c$$

9 b)

$$\int 8\cosh(3 - 4x)\,dx = -2\sinh(3 - 4x) + c$$

9 c)

$$\int 2x\operatorname{sech}^2 x^2\,dx = \tanh x^2 + c$$

9 d)

$$\int x\sinh(3x^2 - 2)\,dx = \frac{1}{6}\cosh(3x^2 - 2) + c$$

9 e)

$$\begin{aligned}\int \frac{\cosh\sqrt{x}}{2\sqrt{x}}\,dx &= \int \frac{1}{2}x^{-1/2}\cosh x^{1/2}\,dx \\ &= \sinh\sqrt{x} + c.\end{aligned}$$

9 f)

$$\begin{aligned}\int \frac{\operatorname{sech}^2(3/x^2)}{4x^3}\,dx &= \int \frac{1}{4}x^{-3}\operatorname{sech}^2(3x^{-2})\,dx \\ &= \frac{1}{24}\tanh\left(\frac{3}{x^2}\right) + c.\end{aligned}$$

10)

$$\begin{aligned}\int_0^{\ln 2} \sinh\left(\frac{3x+\ln 4}{2}\right)\,dx &= \left[\frac{2}{3}\cosh\left(\frac{3x+\ln 4}{2}\right)\right]_0^{\ln 3} \\ &= \frac{2}{3}\cosh\left(\frac{3\ln 2+\ln 4}{2}\right) - \frac{2}{3}\cosh\left(\frac{\ln 4}{2}\right) \\ &= \frac{2}{3}\cosh\left(\frac{1}{2}\ln 32\right) - \frac{2}{3}\cosh(\ln 2) \\ &= \frac{2}{3}\cosh(\ln(4\sqrt{2})) - \frac{2}{3}\cosh(\ln 2) \\ &= \frac{1}{3}\left(4\sqrt{2} + \frac{1}{4\sqrt{2}}\right) - \frac{1}{3}\left(2+\frac{1}{2}\right) \\ &= \frac{11\sqrt{2}}{8} - \frac{5}{6}.\end{aligned}$$

11)

$$\begin{aligned}\int_0^{\sqrt{\ln 2}} 256x\cosh(4x^2)\,dx &= \left[32\sinh(4x^2)\right]_0^{\sqrt{\ln 2}} \\ &= 32\sinh(4\ln 2) = 32\sinh(\ln 16) \\ &= 16\left(16 - \frac{1}{16}\right) = 255.\end{aligned}$$

12)

$$\begin{aligned}\int_0^{\pi/2} \cos x\sinh(\sin x)\,dx &= \left[\cosh(\sin x)\right]_0^{\pi/2} \\ &= \cosh 1 - \cosh 0 = \frac{e+e^{-1}}{2} - 1 \\ &= \frac{e^2-2e+1}{2e} = \frac{(e-1)^2}{2e}.\end{aligned}$$

13 a)

$$\int \tanh x \, dx = \int \frac{\sinh x}{\cosh x} \, dx = \ln |\cosh x| + c$$

13 b)

$$\int \coth x \, dx = \int \frac{\cosh x}{\sinh x} \, dx = \ln |\sinh x| + c$$

14)

$$\begin{aligned}
\int_{\ln 2}^{\ln 4} (2\tanh 2x + 2\coth 2x)\, dx &= [\ln|\cosh 2x| + \ln|\sinh 2x|]_{\ln 2}^{\ln 4} \\
&= \left(\ln \frac{16 + 1/16}{2} + \ln \frac{16 - 1/16}{2}\right) - \left(\ln \frac{4 + 1/4}{2} + \ln \frac{4 - 1/4}{2}\right) \\
&= \ln \frac{257}{32} + \ln \frac{255}{32} - \ln \frac{17}{8} - \ln \frac{15}{8} \\
&= \ln \frac{257}{16}.
\end{aligned}$$

So $p = 257$ and $q = 16$.

15 a)

$$\begin{aligned}
\frac{d}{dx}(\operatorname{cosech} x) &= \frac{d}{dx}((\sinh x)^{-1}) \\
&= -\cosh x (\sinh x)^{-2} \\
&= -\frac{\cosh x}{\sinh^2 x} = -\frac{1}{\sinh x}\frac{\cosh x}{\sinh x} \\
&= -\operatorname{cosech} x \coth x.
\end{aligned}$$

15 b)

$$\int \operatorname{cosech} x \coth x \, dx = -\operatorname{cosech} x + c$$

16 a)

$$\begin{aligned}
\frac{d}{dx}(\operatorname{sech} x) &= \frac{d}{dx}((\cosh x)^{-1}) \\
&= -\sinh x (\cosh x)^{-2} = -\operatorname{sech} x \tanh x.
\end{aligned}$$

16 b)

$$\int \operatorname{sech} x \tanh x \, dx = -\operatorname{sech} x + c$$

17 a)

$$\begin{aligned}\frac{d}{dx}(\coth x) &= \frac{d}{dx}((\tanh x)^{-1}) \\ &= -\operatorname{sech}^2 x(\tanh x)^{-2} \\ &= -\frac{1}{\cosh^2 x}\frac{\cosh^2 x}{\sinh^2 x} = -\frac{1}{\sinh^2 x} = -\operatorname{cosech}^2 x.\end{aligned}$$

17 b)

$$\int \operatorname{cosech}^2 x\, dx = -\coth x + c$$

18 a)

Use Integration by Parts:

$$u = x, \quad \frac{dv}{dx} = \sinh 3x, \quad \frac{du}{dx} = 1, \quad v = \frac{1}{3}\cosh 3x.$$

$$\begin{aligned}\int x \sinh 3x\, dx &= \frac{1}{3}x\cosh 3x - \int \frac{1}{3}\cosh 3x\, dx \\ &= \frac{1}{3}x\cosh 3x - \frac{1}{9}\sinh 3x + c.\end{aligned}$$

18 b)

Use Integration by Parts:

$$u = x^2, \quad \frac{dv}{dx} = \cosh\frac{x}{4}, \quad \frac{du}{dx} = 2x, \quad v = 4\sinh\frac{x}{4}.$$

$$\int x^2 \cosh\frac{x}{4}\, dx = 4x^2\sinh\frac{x}{4} - \int 8x\sinh\frac{x}{4}\, dx.$$

Use Integration by Parts again with $u = 8x$, $dv/dx = \sinh(x/4)$, $du/dx = 8$ and $v = 4\cosh(x/4)$:

$$\begin{aligned}\int x^2 \cosh\frac{x}{4}\, dx &= 4x^2\sinh\frac{x}{4} - \left(32x\cosh\frac{x}{4} - \int 32\cosh\frac{x}{4}\, dx\right) \\ &= 4x^2\sinh\frac{x}{4} - 32x\cosh\frac{x}{4} + 128\sinh\frac{x}{4} + c.\end{aligned}$$

19)

Use Integration by Parts with $u = 9(x+2)$, $dv/dx = \sinh 3x$, $du/dx = 9$ and $v = \frac{1}{3}\cosh 3x$:

$$\begin{aligned}y &= 3(x+2)\cosh 3x - \int 3\cosh 3x\, dx \\ &= 3(x+2)\cosh 3x - \sinh 3x + c.\end{aligned}$$

The curve passes through $(0, 8)$:

$$8 = 3(0+2)\cosh 0 - \sinh 0 + c = 6 + c,$$

so $c = 2$. Therefore

$$y = (3x+6)\cosh 3x - \sinh 3x + 2.$$

20 a)

$$\begin{aligned}\int e^x \sinh x \, dx &= \int e^x \frac{e^x - e^{-x}}{2} \, dx \\ &= \frac{1}{2}\int (e^{2x} - 1) \, dx = \frac{1}{4}e^{2x} - \frac{1}{2}x + c.\end{aligned}$$

20 b)

$$\begin{aligned}\int e^{2x} \cosh x \, dx &= \int e^{2x} \frac{e^x + e^{-x}}{2} \, dx \\ &= \frac{1}{2}\int (e^{3x} + e^x) \, dx = \frac{1}{6}e^{3x} + \frac{1}{2}e^x + c.\end{aligned}$$

20 c)

$$\begin{aligned}\int e^{3x} \sinh 2x \, dx &= \int e^{3x} \frac{e^{2x} - e^{-2x}}{2} \, dx \\ &= \frac{1}{2}\int (e^{5x} - e^x) \, dx = \frac{1}{10}e^{5x} - \frac{1}{2}e^x + c.\end{aligned}$$

20 d)

$$\begin{aligned}\int e^{-4x} \cosh 4x \, dx &= \int e^{-4x} \frac{e^{4x} + e^{-4x}}{2} \, dx \\ &= \frac{1}{2}\int (1 + e^{-8x}) \, dx = \frac{1}{2}x - \frac{1}{16}e^{-8x} + c.\end{aligned}$$

21)

$$\begin{aligned}\cosh 2A &\equiv 2\cosh^2 A - 1 \\ \Rightarrow \cosh^2 A &= \frac{1}{2}\cosh 2A + \frac{1}{2} \\ \therefore \int \cosh^2 x \, dx &= \int \left(\frac{1}{2}\cosh 2x + \frac{1}{2}\right) dx \\ &= \frac{1}{4}\sinh 2x + \frac{1}{2}x + c.\end{aligned}$$

22)

$$\begin{aligned}\cosh 2A &\equiv 1 + 2\sinh^2 A \\ \Rightarrow \sinh^2 A &= \frac{1}{2}\cosh 2A - \frac{1}{2} \\ \therefore \int \sinh^2 6x \, dx &= \int \left(\frac{1}{2}\cosh 12x - \frac{1}{2}\right) dx \\ &= \frac{1}{24}\sinh 12x - \frac{1}{2}x + c.\end{aligned}$$

23)

$$\cosh 2A \equiv 2\cosh^2 A - 1, \qquad \cosh^2 A = \frac{1}{2}(\cosh 2A + 1),$$

$$\begin{aligned}\int_{\ln(5/6)}^{\ln(6/5)} \cosh^2 \frac{x}{2}\, dx &= \frac{1}{2}\int_{\ln(5/6)}^{\ln(6/5)} (\cosh x + 1)\, dx \\ &= \frac{1}{2}[\sinh x + x]_{\ln(5/6)}^{\ln(6/5)} \\ &= \frac{1}{2}\left(\frac{11}{60} + \ln\frac{6}{5} - \left(-\frac{11}{60} + \ln\frac{5}{6}\right)\right) \\ &= \frac{11}{60} + \frac{1}{2}\ln\frac{36}{25} = \frac{11}{60} + \ln\frac{6}{5}.\end{aligned}$$

Thus $p = \frac{11}{60}$ and $q = \frac{6}{5}$.

24)

$$\cosh 2A \equiv 2\cosh^2 A - 1, \qquad \cosh^2 A = \frac{1}{2}(\cosh 2A + 1),$$

$$\begin{aligned}\int_0^{\ln 3} (4\cosh^2 x - 3\cosh x)\, dx &= \int_0^{\ln 3} (2\cosh 2x + 2 - 3\cosh x)\, dx \\ &= [\sinh 2x + 2x - 3\sinh x]_0^{\ln 3} \\ &= \sinh(2\ln 3) + 2\ln 3 - 3\sinh(\ln 3) \\ &= \frac{1}{2}\left(9 - \frac{1}{9}\right) + \ln 9 - \frac{3}{2}\left(3 - \frac{1}{3}\right) \\ &= \frac{40}{9} + \ln 9 - 4 = \frac{4}{9} + \ln 9.\end{aligned}$$

25 a)

$$\begin{aligned}\int \sinh^3 x\, dx &= \int \sinh^2 x\, \sinh x\, dx \\ &= \int (\cosh^2 x - 1)\sinh x\, dx \\ &= \frac{1}{3}\cosh^3 x - \cosh x + c.\end{aligned}$$

Alternatively,

$$\begin{aligned}\sinh^3 x &= \frac{1}{8}(e^x - e^{-x})^3 \\ &= \frac{1}{8}(e^{3x} - 3e^x + 3e^{-x} - e^{-3x}) \\ &= \frac{1}{4}\sinh 3x - \frac{3}{4}\sinh x,\end{aligned}$$

so

$$\int \sinh^3 x\, dx = \frac{1}{12}\cosh 3x - \frac{3}{4}\cosh x + c.$$

25 b)

$$\begin{aligned}\int \cosh^3 x\, dx &= \int \cosh^2 x\, \cosh x\, dx \\ &= \int (\sinh^2 x + 1) \cosh x\, dx \\ &= \frac{1}{3}\sinh^3 x + \sinh x + c.\end{aligned}$$

Alternatively,

$$\begin{aligned}\cosh^3 x &= \frac{1}{8}(e^x + e^{-x})^3 \\ &= \frac{1}{8}(e^{3x} + 3e^x + 3e^{-x} + e^{-3x}) \\ &= \frac{1}{4}\cosh 3x + \frac{3}{4}\cosh x,\end{aligned}$$

so

$$\int \cosh^3 x\, dx = \frac{1}{12}\sinh 3x + \frac{3}{4}\sinh x + c.$$